

An Explicit Exponential Riesz Basis for the Boundary of a Square

Candidate full solution packet for arXiv:2309.14625

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Status. Candidate full affirmative solution; likely valid; novelty plausible but not certified; high-priority human review recommended.

1 Source question

Lai and Sheynis ask on page 3 of [1]:

Does the boundary of the unit square admit a Riesz basis of exponentials?

Their Theorem 1.5 excludes a natural structured class, namely finite unions of translates of the horizontal lattice $\mathbb{Z} \times \{0\}$, but leaves the general question open.

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of the domain. If such connection exists and we know that the boundary of the disk has no frame of exponentials, there may be an answer to the widely open problem that if the unit disk admits a Riesz basis of exponentials. Initiated by this question and the question asking the circle seems difficult, we investigate the following simpler but natural question.

(Qu): Does the boundary of the unit square admit a Riesz basis of exponentials?

Unfortunately, we are not able to obtain an affirmative answer to the proposed question. However, we observe that boundary of a unit square, can be written as a union of two fundamental domains of the closed subgroup $\mathbb{Z} \times \mathbb{R}$ (See Figure [1](#)). This leads us to generalizing

Original question and immediate context, arXiv:2309.14625v1, PDF page 3.

2 Result

Let $Q = [0, 1]^2$, let σ be unnormalized arclength measure on ∂Q , and write $e_{(p,q)}(x, y) = e^{2\pi i(px+qy)}$. Define four disjoint frequency families

$$\begin{aligned}\Lambda_1 &= \{(n, n + \tfrac{1}{2}) : n \in \mathbb{Z}\}, & \Lambda_2 &= \{(n, n + 1) : n \in \mathbb{Z}\}, \\ \Lambda_3 &= \{(n - \tfrac{1}{2}, -n + \tfrac{1}{2}) : n \in \mathbb{Z}\}, & \Lambda_4 &= \{(n - \tfrac{1}{2}, -n) : n \in \mathbb{Z}\},\end{aligned}$$

and $\Lambda = \bigcup_{j=1}^4 \Lambda_j$.

Theorem 1. *The exponential system $E(\Lambda) = \{e_\lambda : \lambda \in \Lambda\}$ is a Riesz basis of $L^2(\partial Q, \sigma)$. Its optimal squared Riesz bounds are*

$$4 - 2\sqrt{2} \quad \text{and} \quad 4 + 2\sqrt{2}.$$

Equivalently, for every finitely supported scalar family (c_λ) ,

$$(4 - 2\sqrt{2}) \sum_{\lambda \in \Lambda} |c_\lambda|^2 \leq \left\| \sum_{\lambda \in \Lambda} c_\lambda e_\lambda \right\|_{L^2(\partial Q)}^2 \leq (4 + 2\sqrt{2}) \sum_{\lambda \in \Lambda} |c_\lambda|^2,$$

and the closed span is all of $L^2(\partial Q, \sigma)$.

Thus the source question has an affirmative answer. The set Λ is not in the structured horizontal-coset class excluded in [1]: its four families lie on two lines of slope 1 and two lines of slope -1 .

3 Proof intuition

Opposite edges of the square provide symmetric and antisymmetric channels. An integral coordinate makes the corresponding opposite-edge values agree, while a half-integral coordinate makes them disagree. The four frequency families use all four integral/half-integral phase combinations, and the two negative-slope families introduce exactly one reflection of the interval.

After grouping coefficients into four ordinary Fourier series, all apparently infinite-dimensional coupling consists only of unimodular multiplication and reflection, hence unitary operations. Removing those unitary operations leaves one constant 4×4 matrix. Its determinant is -2 , which gives completeness, and its two singular values give the exact Riesz bounds. The integer offset $n + 1$ in Λ_2 is essential: it changes one channel by a full Fourier shift and prevents the endpoint degeneracy seen in simpler parallel-coset candidates.

4 Proof

Parametrize the four edges, in the order bottom, top, left, right, by

$$(t, 0), \quad (t, 1), \quad (0, t), \quad (1, t), \quad 0 < t < 1.$$

This identifies $L^2(\partial Q, \sigma)$ unitarily with $L^2(0, 1)^4$. The restriction of an exponential of frequency (p, q) is

$$(e_p, e^{2\pi i q} e_p, e_q, e^{2\pi i p} e_q), \quad e_r(t) = e^{2\pi i r t}. \quad (1)$$

For a finitely supported coefficient family, write its coefficients on Λ_j as $(c_{j,n})_{n \in \mathbb{Z}}$ and form the ordinary Fourier series

$$C_j(t) = \sum_{n \in \mathbb{Z}} c_{j,n} e^{2\pi i n t}.$$

By Parseval,

$$\sum_{j=1}^4 \sum_{n \in \mathbb{Z}} |c_{j,n}|^2 = \sum_{j=1}^4 \|C_j\|_2^2. \quad (2)$$

Set $s(t) = e^{\pi i t}$, so $s^2(t) = e^{2\pi i t}$, and let

$$(Rf)(t) = f(1 - t), \quad 0 < t < 1.$$

Both multiplication by s and R are unitary on $L^2(0, 1)$, and

$$R(sf) = -s^{-1}Rf. \quad (3)$$

Pass from opposite-edge values (f_b, f_t, f_l, f_r) to the unitary channel coordinates

$$h_+ = \frac{f_b + f_t}{\sqrt{2}}, \quad h_- = \frac{f_b - f_t}{\sqrt{2}}, \quad v_+ = \frac{f_l + f_r}{\sqrt{2}}, \quad v_- = \frac{f_l - f_r}{\sqrt{2}}.$$

Using (1) for the four displayed families gives, directly,

$$h_- = \sqrt{2}(C_1 + s^{-1}C_3), \quad h_+ = \sqrt{2}(C_2 + s^{-1}C_4), \quad (4)$$

$$v_+ = \sqrt{2}(sC_1 + s^2C_2), \quad v_- = \sqrt{2}(sRC_3 + RC_4). \quad (5)$$

For example, Λ_1 is horizontally antisymmetric and vertically symmetric, while Λ_4 is horizontally symmetric and vertically antisymmetric. The negative sign in its second coordinate accounts for the reflection R .

Now make the unitary change of input variables

$$a = C_1, \quad b = sC_2, \quad c = s^{-1}C_3, \quad d = C_4.$$

On the output, multiply h_+ by s , multiply v_+ by s^{-1} , and apply R to v_- . These are again unitary operations. Equations (4)–(5), together with (3), become

$$\frac{1}{\sqrt{2}} \begin{pmatrix} h_- \\ sh_+ \\ s^{-1}v_+ \\ Rv_- \end{pmatrix} = M \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Thus the entire synthesis operator is unitarily equivalent to pointwise multiplication by the constant matrix $\sqrt{2}M$.

A direct calculation gives

$$\det M = -2, \quad \det(xI - M^*M) = (x^2 - 4x + 2)^2.$$

Hence the eigenvalues of M^*M are $2 - \sqrt{2}$ and $2 + \sqrt{2}$, each with multiplicity two. Combining this with (2) proves the two Riesz inequalities, with squared bounds $2(2 \mp \sqrt{2}) = 4 \mp 2\sqrt{2}$.

Finally, M is invertible; explicitly,

$$M^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & 1 \end{pmatrix}.$$

All the transformations before and after M are unitary. Therefore the synthesis operator from $\ell^2(\mathbb{Z})^4$ onto $L^2(\partial Q, \sigma)$ is a bounded bijection. This proves completeness and the Riesz-basis assertion. Since constant eigenvectors of M^*M , multiplied by arbitrary nonzero L^2 functions, realize the extremal operator norms (and finite Fourier series approximate them), the stated bounds are optimal. $\square$

Corollary 2. *Every Euclidean square, with arclength measure on its boundary, admits an explicit Riesz basis of exponentials.*

Proof. A translation only multiplies each restricted exponential by a unimodular constant. If the square is the image of Q under $x \mapsto c + \ell O x$, with O orthogonal and $\ell > 0$, use the transformed frequency set $\ell^{-1}O\Lambda$. Pullback multiplies arclength by ℓ , so the Riesz bounds scale by ℓ and the basis property is unchanged. $\square$

5 Verification and novelty audit

The reusable script `code/verify_basis.py` checks $\det M = -2$, the exact characteristic polynomial $(x^2 - 4x + 2)^2$, the displayed inverse, and forty randomized direct syntheses on the four original edge restrictions. The code is a sanity check; the matrix conjugation above is the proof.

A bounded search on 17 August 2026 covered the exact question and title, arXiv:2309.14625, “square boundary” with “Riesz basis of exponentials”, the authors’ names, close polygon/line-segment terminology, citation-style queries, and arXiv results through that date. Kolountzakis and Lai prove that closed polygonal lines, including the square boundary, are not *spectral* [2]; that excludes an orthonormal exponential basis, not a Riesz basis. Lai’s 2025 open-problems slides still present the Riesz-basis issue as open [3]. No later affirmative or negative Riesz-basis answer, and no matching four-family construction, was found. Novelty is therefore plausible, not certified, and no priority claim is made.

6 Scope and reviewer focus

The theorem answers the square-boundary question but not the analogous plus-space question in [1]. It does not contradict the source paper’s negative theorem because Λ is not a finite union of translates of $\mathbb{Z} \times \{0\}$. Review should focus first on the four channel identities (4)–(5), especially the full shift in Λ_2 and the reflection identity (3); after those, the proof is finite-dimensional exact linear algebra.

References

- [1] C.-K. Lai and A. Sheynis, *Riesz bases of exponentials for multi-tiling measures*, Sampling Theory, Signal Processing, and Data Analysis 21 (2023), Article 29; arXiv:2309.14625.
- [2] M. N. Kolountzakis and C.-K. Lai, *Non-spectrality of some piecewise smooth curves and unions of line segments*, arXiv:2507.00581 (2025).
- [3] C.-K. Lai, *Some Open Problems about Singular Spectral Measures*, ICTFB 2025 invited lecture slides, Xi’an, 2025.